

Unplugged Activity

Parallel Computing

Unplugged Parallel Computing Flag of Mauritius ([Mauritius - Wikipedia](#))

Shared Memory: Cases

Materials needed: 8x11 pieces of paper with grids and 4 markers and crayons of each color.

Tell the students that the cells must be completely filled in.

Each stripe = 2 rows (8 cells per row), processor(s) will color completely each cell in row(s) and/or column(s)

Case 1: (1 P or sequential)

Student 1: Processor colors first stripe (cell by cell), second stripe (cell by cell), third stripe (cell by cell), and fourth stripe (cell by cell)

Student 2: times Student 1

P1								

Case 2: (2 P)

Divide by colors

Student 1: Processor 1 colors red and blue stripes (cell by cell)

Student 2: Processor 2 colors yellow and green stripes (cell by cell)

Student 3: records the time of the slower processor

P1	1	2	3	4	5	6	7	8
	9	10	11	12	13	14	15	16
	17	18	19	20	21	22	23	24
	25	26	27	28	29	30	31	32
P2	1	2	3	4	5	6	7	8
	9	10	11	12	13	14	15	16
	17	18	19	20	21	22	23	24
	25	26	27	28	29	30	31	32

Case 3: (4 P)

Divide by colors

Student 1: Processor 1 colors first stripe red (cell by cell)

Student 2: Processor 2 colors second stripe blue (cell by cell)

Student 3: Processor 3 colors third stripe yellow (cell by cell)

Student 4: Processor 4 colors fourth stripe green (cell by cell)

Student 5: records the time of the slower processor

P1	1	2	3	4	5	6	7	8
	9	10	11	12	13	14	15	16
P2	1	2	3	4	5	6	7	8
	9	10	11	12	13	14	15	16
P3	1	2	3	4	5	6	7	8
	9	10	11	12	13	14	15	16
P4	1	2	3	4	5	6	7	8
	9	10	11	12	13	14	15	16

Case 4: (4 P)

Student 1: Processor 1 colors the first 2 columns cell by cell (all markers are needed)

Student 2: Processor 2 colors second 2 columns cell by cell (all markers are needed)

Student 3: Processor 3 colors third 2 columns cell by cell

Student 4: Processor 4 colors fourth 2 columns cell by cell

Student 5: records the time of the slower processor

P1		P2		P3		P4	
1	9	1	9	1	9	1	9
2	10	2	10	2	10	2	10
3	11	3	11	3	11	3	11
4	12	4	12	4	12	4	12
5	13	5	13	5	13	5	13
6	14	6	14	6	14	6	14
7	15	7	15	7	15	7	15
8	16	8	16	8	16	8	16